3. Functional Requirements

Functional requirements define what the system must do — the capabilities the platform must provide to support business processes and customer interactions. Requirements are prioritized using the MoSCoW framework: Must (non-negotiable), Should (high priority), Could (desirable if capacity allows), and Won't (explicitly excluded from this release).

ID	Requirement Description	Priority	Category
FR-01	The system must synchronize inventory levels across all regional warehouses and e-commerce channels in real time, with updates propagated within 5 seconds of a stock-level change	Must	Inventory Consistency
FR-02	Checkout and product-browsing services must scale automatically in response to demand spikes, provisioning additional compute capacity without manual intervention	Must	Scalability
FR-03	The platform must expose RESTful APIs to allow authorized third-party logistics, payment, and marketplace partners to integrate with order management and inventory systems	Must	Integration
FR-04	Orders placed in any region must be routable to the nearest available fulfilment centre, with automatic failover to an alternative centre if primary capacity is unavailable	Must	Order Routing
FR-05	The platform must provide real-time product availability status on all customer-facing channels, reflecting warehouse and in-store stock levels within the same synchronization window	Should	Availability Display
FR-06	The system should support multi-currency and multi-language checkout flows, enabling the platform to onboard new regional markets without core code changes	Should	Localization
FR-07	An event-driven notification service could alert store and warehouse managers when stock levels fall below configurable thresholds for high-velocity SKUs	Could	Alerting

4. Non-Functional Requirements

Non-functional requirements define system qualities, operational constraints, and performance expectations. Each requirement includes a measurable target to enable objective validation during testing and acceptance. Priorities follow the same MoSCoW framework applied to functional requirements.

ID	Requirement Description	Measurable Target	Priority	Category
NFR-01	The platform must remain continuously available during peak seasonal trading events, including flash sales and holiday campaigns	99.95% uptime per calendar month (≤ 4.4 hrs downtime/month)	Must	Availability
NFR-02	End-to-end checkout transaction latency, from cart submission to order confirmation, must meet defined thresholds to protect conversion rates	≤ 500ms at p95 under peak load	Must	Performance
NFR-03	The platform must define and enforce recovery objectives to ensure that service disruptions can be detected and resolved within acceptable business impact windows	RTO ≤ 1 hour; RPO ≤ 15 minutes	Must	Resilience
NFR-04	All customer personal data, payment information, and order records must be encrypted both in transit and at rest using current industry-accepted standards	TLS 1.3 in transit; AES-256 at rest	Must	Security
NFR-05	Customer and transaction data must be stored and processed within the geographic boundary of the customer's originating region to satisfy applicable data residency regulations	Compliance with GDPR (EU), PDPA (APAC), and applicable regional frameworks	Must	Compliance
NFR-06	The platform architecture must support independent horizontal scaling of individual service components without requiring full-stack redeployment or scheduled downtime	Linear scaling to 10× baseline traffic with ≤ 10% latency degradation	Should	Scalability
NFR-07	Comprehensive telemetry, including service health metrics, error rates, and latency distributions, should be observable through a centralized monitoring dashboard	100% of critical service paths instrumented; p99 latency visible in dashboard	Should	Observability

5. Requirements Review Checklist

The following criteria were applied when reviewing this document for quality and completeness:

✓ **Clarity:** Each requirement is written as a single, unambiguous statement that can be independently validated

✓ **Separation:** Functional requirements describe system capabilities (what it does); non-functional requirements describe system qualities (how well it does it)

✓ **Priority Levels:** Every requirement includes a MoSCoW priority to support scope management and trade-off decisions

✓ **Measurable Targets:** All non-functional requirements include a quantified, testable target (e.g., latency thresholds, uptime percentages, recovery objectives)

✓ **Stakeholder Coverage:** Each functional and non-functional requirement traces back to at least one identified stakeholder concern or business objective